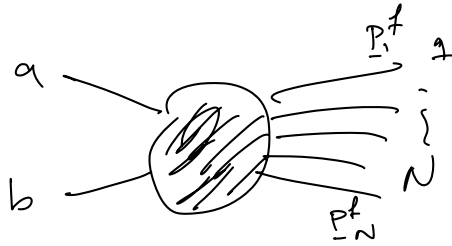


$$\frac{dN_r}{dt} = \sigma \frac{dN_b}{dt} n_b \cdot d$$

$\hookrightarrow$ density of target

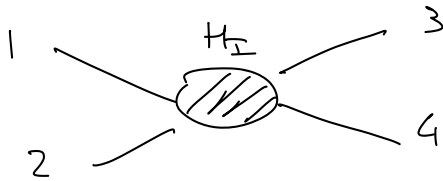


$|i\rangle \quad |f\rangle$
 $a+b \rightarrow 1+\dots+N$
 scattering

$$\sigma = \frac{\mathcal{S}}{4 \sqrt{(\underline{p}_a \cdot \underline{p}_b)^2 - m_a^2 m_b^2}} \int |\mathcal{M}|^2 (2\pi)^4 \delta(\underline{p}_a + \underline{p}_b - \sum_i \underline{p}_i^f) \times$$

$$\times \prod_{j=1}^N \frac{1}{2 \sqrt{\underline{p}_j^2 + m_j^2}} \frac{d^3 \underline{p}_j}{(2\pi)^3}$$

$a+b \rightarrow c+d$ $\mathcal{S} = \frac{1}{2!} \frac{1}{3!}$

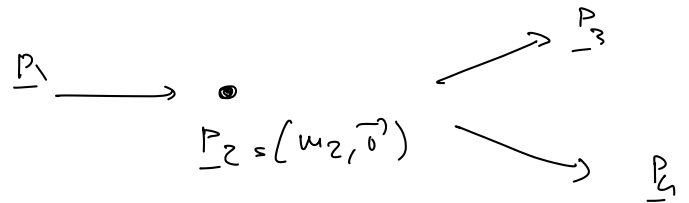
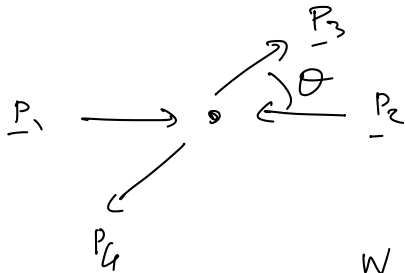


Example: $1+2 \rightarrow 3+4$

$a+b \rightarrow 3+4$

Center of Mass

Fixed target: 2 at rest.



center of mass

$$\sum_{i=1}^N \vec{p}_i^{\text{in}} = 0 \Rightarrow \vec{p}_1 + \vec{p}_2 = 0$$

$$\sigma = \frac{1}{4 \sqrt{(\underline{p}_1 \cdot \underline{p}_2)^2 - m_1^2 m_2^2}} \int |\mathcal{M}|^2 \delta^4(\underline{p}_1 + \underline{p}_2 - \underline{p}_3 - \underline{p}_4) (2\pi)^4 \frac{1}{4 (2\pi)^6} \frac{d^3 \underline{p}_3}{\sqrt{\underline{p}_3^2 + m_3^2}} \frac{d^3 \underline{p}_4}{\sqrt{\underline{p}_4^2 + m_4^2}}$$

$$\vec{p}_1 = -\vec{p}_2 = \vec{p}_{\text{in}}$$

$$\vec{p}_3 = -\vec{p}_4 = \vec{p}_{\text{out}}$$

$$\delta(E_1 + E_2 - E_3 - E_4) \delta^3(\underbrace{\vec{p}_1 + \vec{p}_2}_{=0} - \vec{p}_3 - \vec{p}_4) \hookrightarrow \vec{p}_3 = -\vec{p}_4$$

$$\sigma = \frac{1}{4 \sqrt{(\underline{p}_1 \cdot \underline{p}_2)^2 - m_1^2 m_2^2}} \frac{1}{4(2\pi)^2} \int |\mathcal{M}|^2 \delta(E_{in} - E_{fin}) \frac{d^3 \underline{p}_3}{\sqrt{\underline{p}_3^2 + m_3^2} \sqrt{\underline{p}_4^2 + m_4^2}}$$

change of variable

$$u = E_3 + E_4$$

$\delta(E_1 + E_2 - u) \Rightarrow$ same trick used in last lecture

$$d^3 \underline{p}_3 = p_3^2 dp_3 d\Omega$$

$$\frac{du}{dp} = \frac{p}{\sqrt{p^2 + m_3^2}} + \frac{p}{\sqrt{p^2 + m_4^2}} = \frac{p u}{\sqrt{\dots} \sqrt{\dots}}$$

$$\int p^2 dp \delta(u - E_1 - E_2) \underbrace{\frac{1}{\sqrt{\dots} \sqrt{\dots}}}_{\frac{1}{p} \frac{1}{u} \frac{du}{dp}} \int |\mathcal{M}|^2 d\Omega$$

$$\int dp \delta(u - E_1 - E_2) \frac{p^2}{p} \frac{1}{u} \frac{du}{dp} =$$

$$\int \delta(u - E_1 - E_2) \frac{p}{u} du = \frac{p}{E_1 + E_2}$$

$$\sigma = \frac{1}{16\pi^2} \frac{1}{4 \sqrt{(\underline{p}_1 \cdot \underline{p}_2)^2 - m_1^2 m_2^2}} \cdot \frac{|\underline{p}_{out}|}{E_1 + E_2} \int |\mathcal{M}|^2 d\Omega$$

Depends on frame

Center of mass:

$$\underline{p}_1 = (E_1, \underline{p}_{in}) \quad \underline{p}_2 = (E_2, -\underline{p}_{in}) \quad \xrightarrow{\underline{p}_1} \xleftarrow{\underline{p}_2}$$

$$S = (\underline{p}_1 + \underline{p}_2)^2 = (E_1 + E_2)^2 \quad \sqrt{S} = E_1 + E_2$$

$$4 \sqrt{(\underline{p}_1 \cdot \underline{p}_2)^2 - m_1^2 m_2^2} = 4 |\underline{p}_{in}| (E_1 + E_2) = 4 |\underline{p}_{in}| \sqrt{S}$$

$$\frac{d\sigma}{d\Omega} = \frac{1}{16\pi^2} \frac{1}{4 |\underline{p}_{in}| \sqrt{S}} \frac{|\underline{p}_{out}|}{\sqrt{S}} |\mathcal{M}|^2 = \frac{1}{64\pi^2} \frac{1}{S} \frac{|\underline{p}_{out}|}{|\underline{p}_{in}|} |\mathcal{M}|^2$$

Fixed Target Frame

$$\underline{P}_1 = (E_1, \vec{P}_{in})$$

$$\underline{P}_2 = (m_2, \vec{0})$$

$$\vec{P}_{in} = \vec{P}_{out} + \vec{P}_4$$

$$\underline{P}_3 = (E_3, \vec{P}_{out})$$

$$\underline{P}_4 = (E_4, \vec{P}_4)$$

Integral of phase space different before.

$$\delta(E_1 + E_2 - E_3 + E_4) = \delta(E_1 + m_2 - E_3 - E_4)$$

$$\delta(\vec{P}_1 + \vec{P}_2 - \vec{P}_3 - \vec{P}_4) = \delta(\vec{P}_{in} - \vec{P}_{out} - \vec{P}_4)$$

$$4 \sqrt{(\underline{P}_1 \cdot \underline{P}_2)^2 - m_1^2 m_2^2} = 4 \sqrt{(E_1 m_2)^2 - m_1^2 m_2^2} = 4 m_2 \sqrt{E_1^2 - m_1^2}$$

$$= 4 m_2 |\vec{P}_1| = 4 m_2 |\vec{P}_{in}|$$

$$\frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2} \frac{1}{(m_2)^2} \frac{|\vec{P}_{out}|}{|\vec{P}_{in}|} |\mathcal{M}|^2$$

Fixed target experiments:

LAB frame same as fixed target frame

Colliding beam experiments:

if beams with same energy but opposite direction

$\Rightarrow$ LAB $\equiv$ Center of mass.

$\vec{P}_{in}$: Known because energy/momentum of beam

$\vec{P}_{out}$: Can be measured with detectors

$|M|$: to be computed and compare to measurement of σ

Rutherford scattering:

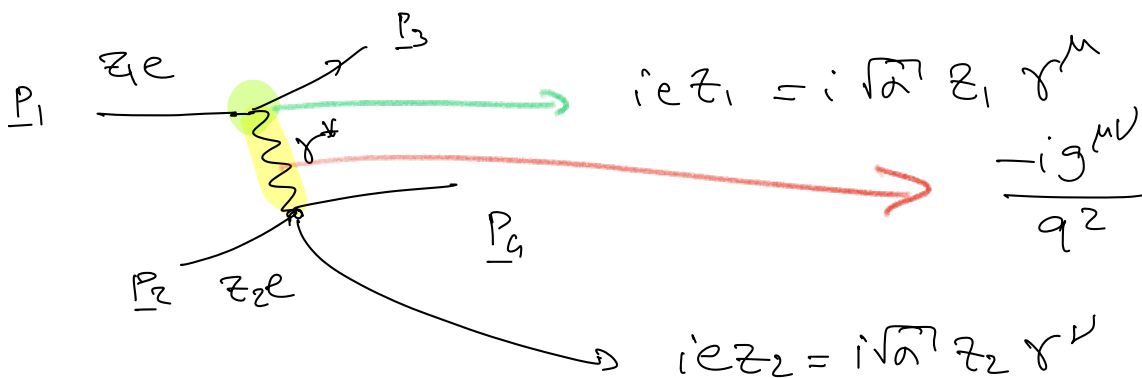
$\alpha + N \rightarrow \alpha + N$
elastic scattering.

$\alpha = {}^4_2\text{He}$ $m_\alpha = 3.7 \text{ GeV}$ $K = 5 \text{ MeV}$

$N: m \approx 80 \text{ GeV} \gg m_\alpha$

$\gamma + e^- \rightarrow \gamma + e^-$ Compton scattering.

EM interaction $e^- + p \rightarrow e^- + p$ $\alpha = \frac{e^2}{4\pi}$



$q = p_1 - p_3$ $p_1 = p_3 + q$ $q = p_1 - p_2$

Rutherford:

$m_1 = m_\alpha$

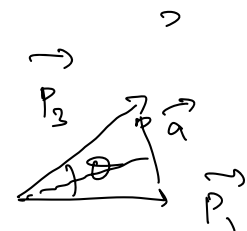
$z_1 = 2$

$\alpha + N \rightarrow \alpha + N$

$m_2 = m_N$

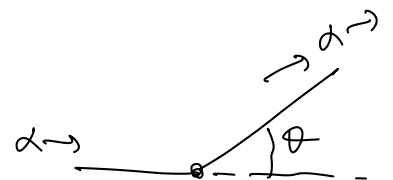
$z_2 = z_N$

$m_N \gg m_\alpha \Rightarrow \vec{p}_N \approx 0$



$|\vec{p}_3| = |\vec{p}_1| = q = 2 p_{in} \sin \frac{\theta}{2}$

$|M|^2 = \left(\frac{\alpha m_N}{q^2} \right)^2 (m_1^2 + p_{in}^2 \cos^2 \frac{\theta}{2})$



$$\mu \sim \propto \frac{1}{q^2} \Rightarrow |M|^2 \sim \alpha^2 \frac{1}{q^4}$$

$$\frac{d\sigma}{d\Omega} \sim \frac{1}{\cancel{m^2}} \left(\frac{\cancel{\alpha m}}{q^2} \right)^2 \underbrace{(m_1^2 + p_{in}^2 \cos^2 \frac{\theta}{2})}_{\text{Mott Formula}}$$

$$q^2 = 4 p_{in}^2 \sin^2 \frac{\theta}{2} = \sigma \sim \frac{1}{p_{in}^4} \frac{1}{\sin^4 \frac{\theta}{2}}$$

$$m_1^2 = E_1^2 - \vec{p}_{in}^2 \Rightarrow \cos^2 \frac{\theta}{2} = 1 - \sin^2 \frac{\theta}{2}$$

$$m_1^2 + p_{in}^2 \left(1 - \sin^2 \frac{\theta}{2} \right) = \underbrace{m_1^2 + p_{in}^2}_{E_1^2} - p_{in}^2 \sin^2 \frac{\theta}{2}$$

$$= E_1^2 \left(1 - \underbrace{\frac{p_{in}^2}{E_1^2}}_{\beta^2} \sin^2 \frac{\theta}{2} \right) \quad \underline{P}_1 = (E_1, \vec{p}_{in})$$

$$= E_1^2 (1 - \beta^2 \sin^2 \frac{\theta}{2})$$

β : of projectile.

Rutherford: $\beta \rightarrow 0$.

$$E = \sqrt{p_{in}^2 + m_1^2} \simeq m_1$$

$$\text{Rutherford} \quad \frac{d\sigma}{d\Omega} \sim \frac{d^2}{(6 p_{in}^4 \sin^4 \frac{\theta}{2})} \frac{|\vec{p}_{out}|}{|\vec{p}_{in}|} m_\alpha^2$$

σ : Lorentz invariant.

$\frac{d\sigma}{d\Omega}$: not Lorentz invariant. $d\Omega = \sin\theta d\theta d\varphi$

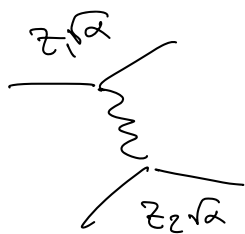
LHC: $P = 6.5 \text{ TeV} \quad \longleftrightarrow \quad P = 6.5 \text{ TeV}$

LAB $\equiv$ Center of mass.

$$e^- + N \rightarrow e^- + N$$

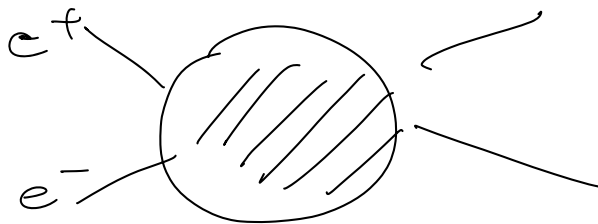
$$\left(\frac{d\sigma}{d\Omega}\right)_{\text{Mott}} = \left(\frac{d\sigma}{d\Omega}\right)_{\text{Ruth.}} (1 - \beta^2 \sin^2 \frac{\theta}{2})$$

$$\frac{d\sigma}{d\Omega} \Big|_{\text{Ruth.}} = (-) \frac{(z_1 z_2 \alpha)^2 |\vec{P}_{\text{out}}|}{q^4 |\vec{P}_{\text{in}}|} E_{\text{in}}^2$$

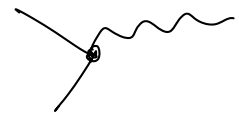


$$\mathcal{M} \propto z_1 z_2 \alpha \Rightarrow |\mathcal{M}|^2 (z_1 z_2 \alpha)^2$$

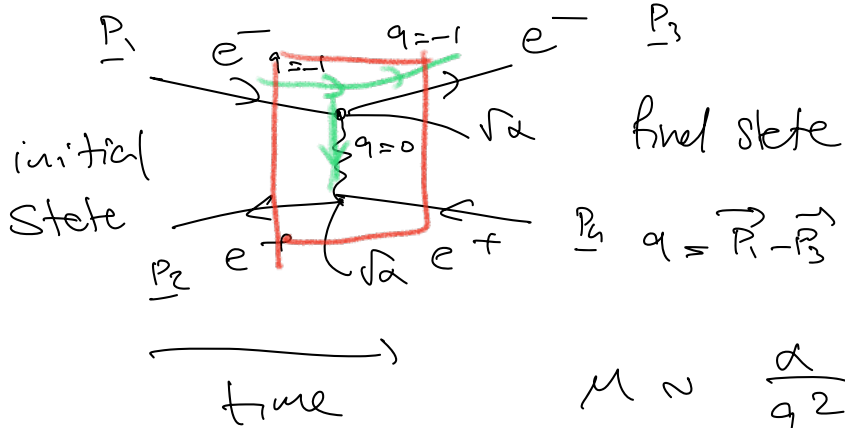
$$e^+ + e^- \rightarrow ? \quad \text{in QED}$$



QED vertex



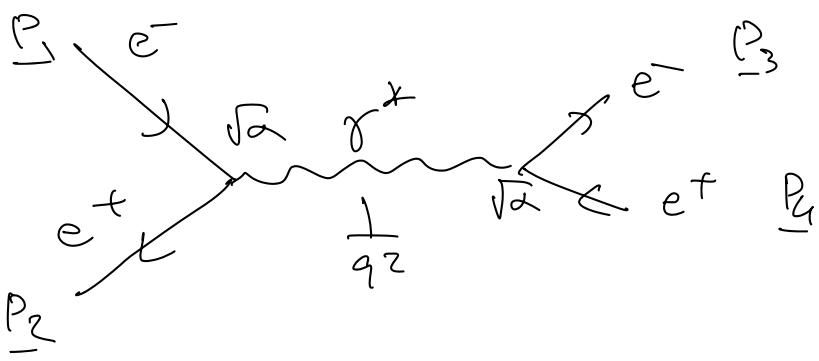
$\mathcal{M}_1$



$$e^+ e^- \rightarrow e^+ e^-$$

t-channel

$\mathcal{M}_2$



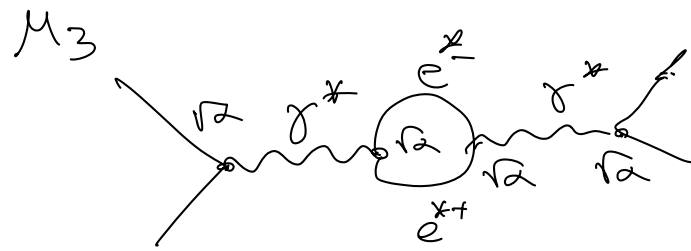
$$\underline{q} = \underline{p}_1 + \underline{p}_2$$

$$q^2 = (\underline{p}_1 + \underline{p}_2)^2 = s$$

$$\mu = \mu_1 + \mu_2$$

$$e^+e^- \rightarrow e^+e^-$$

$$\frac{d\sigma}{d\Omega} \propto |\mathcal{M}|^2$$

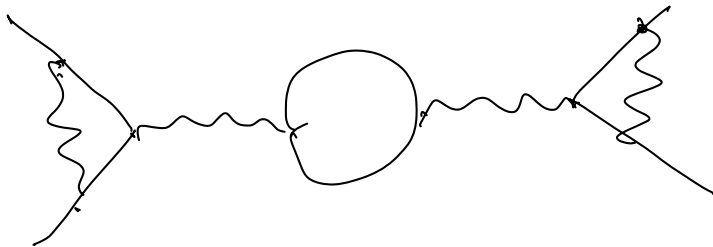


$$\mu_3 \propto \alpha^2$$

$$\alpha = \frac{1}{137}$$

$$\mu = \underbrace{\mu_1 + \mu_2}_{\propto} + \underbrace{\mu_3}_{\propto^2}$$

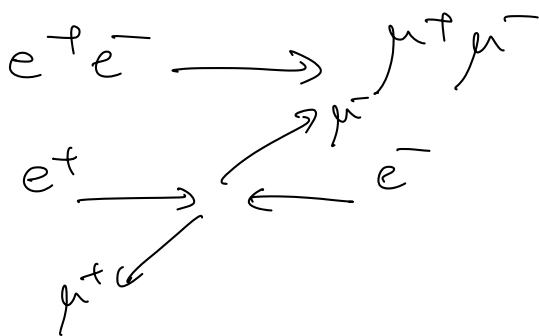
$$1 + 0.01$$



$$\mu \propto \alpha^4$$

$$e^+e^- \rightarrow e^+e^- \quad \text{elastic scattering.}$$

$$\sqrt{s} = \sqrt{\underline{p}_1^2} = \sqrt{(\underline{p}_1 + \underline{p}_2)^2}$$



$$m_e = 0.5 \text{ MeV}$$

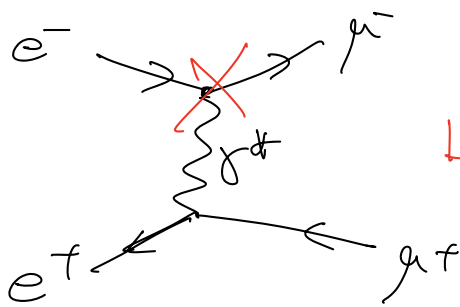
$$m_\mu = 106 \text{ MeV.}$$

$$\sqrt{s} = E_1 + E_2 = 2 E_{\text{beam}} = 2 m_\mu \Rightarrow E_{\text{beam}} > m_\mu$$

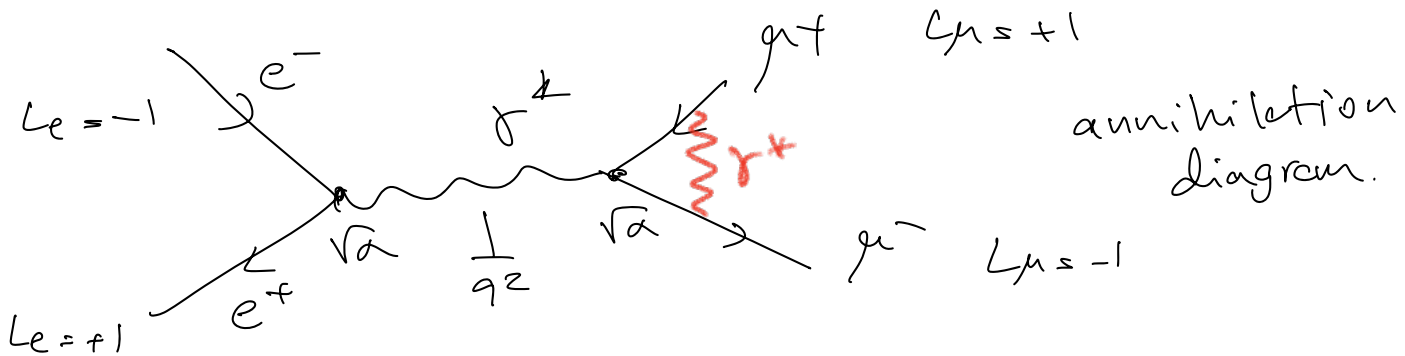
$$E_e \gg 106 \text{ MeV}$$

$$\sqrt{s} < 2m_\mu$$

$$e^+e^- \rightarrow \mu^+\mu^- \quad \sqrt{s} \geq 2m_\mu$$



not in QED
Lepton Flavor not conserved



annihilation diagram.

$$M_{ee \rightarrow \mu\mu} \sim \frac{\alpha}{q^2}$$

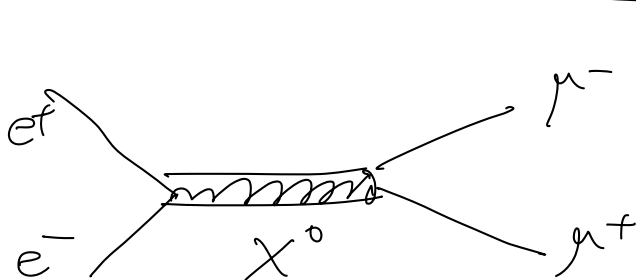
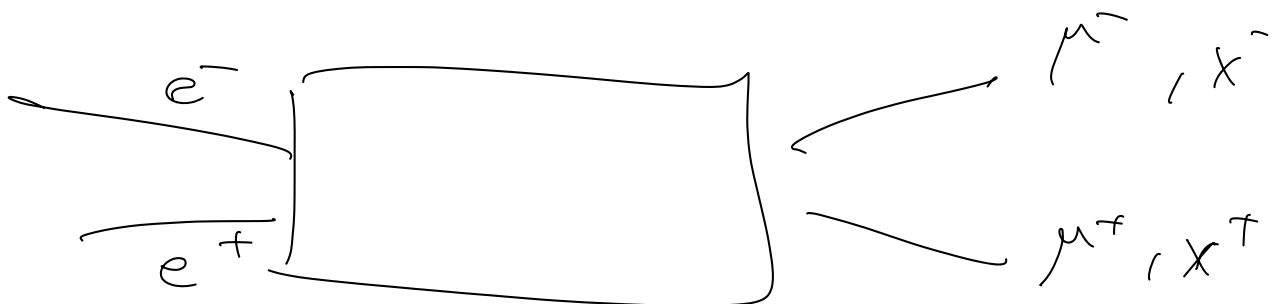
$$q = p_1 + p_2$$

$$|q| = \sqrt{s}$$

$$e^+e^- \rightarrow \chi^+\chi^-$$

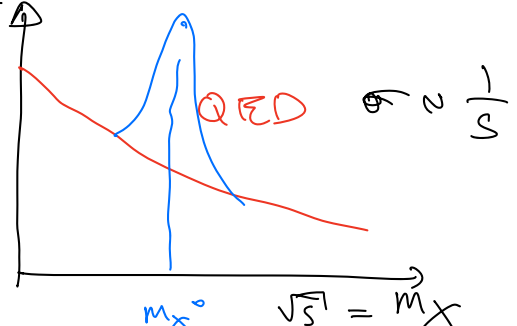
$$\sqrt{s} \geq 2m_\chi$$

~~$$\chi^+\chi^+$$~~
~~$$\chi^0\bar{\chi}^0$$~~
~~$$\chi^0\chi^+$$~~
~~$$\chi^-\chi^0$$~~



$$\sqrt{s} \geq m_\chi$$

event.



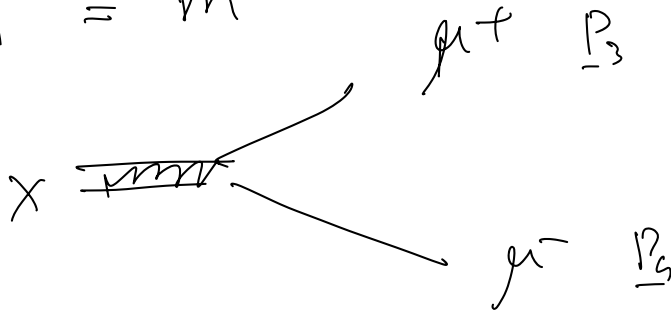
invariant mass: $\sqrt{(\sum_i \underline{p}_i)^2}$

$$(\underline{p}_3 + \underline{p}_4)^2 = M_{\text{inv}}^2$$

In relativity: $\underline{p} = (E, \vec{p})$ $E^2 = m^2 + p^2$

In rest frame of particle: $\underline{p} = (E, \vec{0}) = (m, \vec{0})$

$$\sqrt{|\underline{p}|^2} = m$$



$$(\underline{p}_3 + \underline{p}_4)^2 = m_X^2$$